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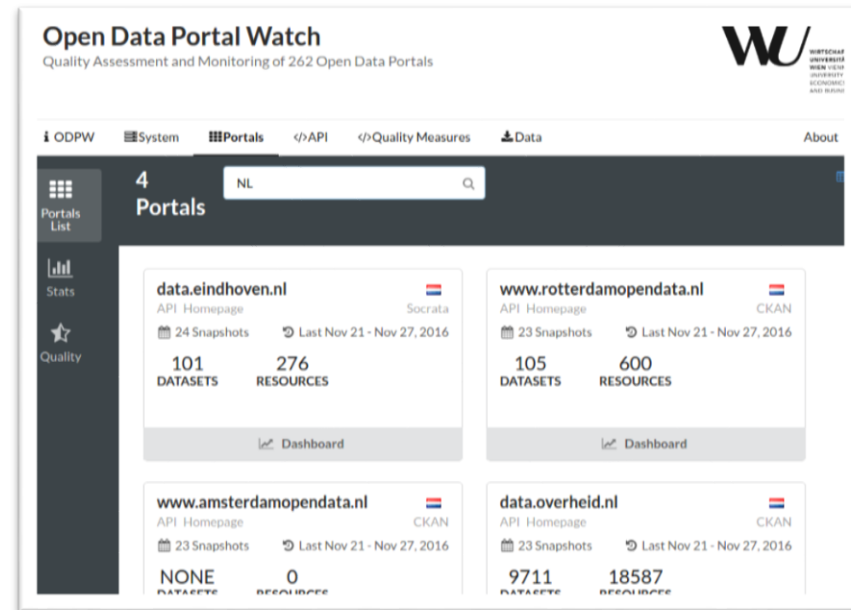
Challenges of mapping current **CKAN** metadata to **DCAT**

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What we do:

- Profiling Open Data portals:
 - **Quality assessment**
 - Evolution monitoring



<http://data.wu.ac.at/portalwatch>

- Different software frameworks:
 - We monitor ~250 portals in total
 - Most of them **CKAN: 133 portals, 514k datasets**
- We harmonize metadata by **mapping** it to **DCAT**

CKAN (mostly) exports DCAT

- CKAN provides **extension** to export DCAT
 - Mapping of *datasets* and *resources* to `dcat:Dataset` and `dcat:Distribution`
 - Recent version of extension supports DCAT-AP
- 93 of 133 CKAN portals provide DCAT export

CKAN provides “extra” keys

- CKAN can include additional metadata keys
 - Added by portal provider, or other CKAN extension

ADDITIONAL INFORMATION

Added to data.gov.uk	04/04/2011
Theme	Environment
Temporal coverage	1990 - 2006
Geographic coverage	England, Wales
Precision	nearest per cent
Update frequency	annual
Temporal granularity	year

What “extra” keys are available?

- 3607 different extra keys in 514k datasets
- Extra keys in multiple portals:

Portals	1	2	3 – 9	10 – 19	≥ 20
Extra keys	2269	1131	172	30	5

- Most frequent extra keys:

Extra key	Datasets	Portals	Origin	DCAT key
harvest_object_id	268241	30	Harvesting extension	—
spatial	245211	33	Spatial extension	dct:spatial
harvest_source_id	243136	29	Harvesting extension	—
harvest_source_title	243043	29	Harvesting extension	—
guid	150811	20	Spatial extension	—
resource-type	148843	15	Spatial extension	—
contact-email	148671	17	Spatial extension	dcat:contactPoint
metadata-date	141758	15	Spatial extension	dct:issued/modified

Current mapping of “extra” keys

3 different cases, depending on version and configuration of CKAN-to-DCAT extension:

- ***Portal-specific mapping:***

Portal defines mapping for metadata key to property, e.g.:

`"temporal_coverage" → dct:temporal`

- ***Generic mapping by extension:***

Pattern for exporting all available extra keys, e.g.:

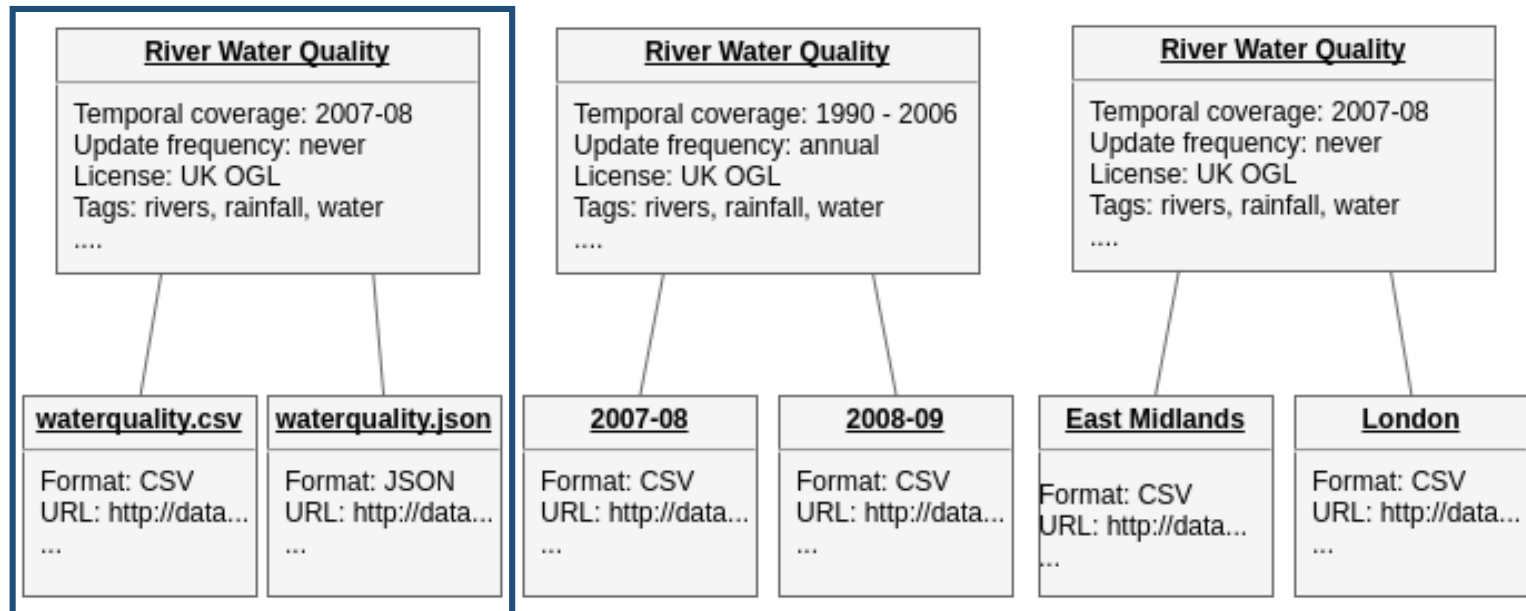
```
dc:relation [  
    rdfs:label "geographic_coverage" ;  
    rdf:value "101000: England, Wales"  
]
```

- ***No mapping:***

Retrieved DCAT description returns no mapping for extra keys

How to model CKAN resources?

- DCAT distribution:
 - *dataset might be available in different forms, these forms might represent different formats or endpoints*
- Use of resources in CKAN:



DCAT distribution

Summary

- *We do:* **monitoring** of ~250 Open Data portals
- *We gained insights:*
 - ... into **current use** of CKAN metadata keys
 - ... into **issues when mapping** these keys to DCAT:
 - Different mapping output for “extra” keys
 - No predefined mapping for frequently used “extra” keys
 - DCAT specification defines distributions as different forms/formats of same content (and not by other dimensions)

Conclusions & Suggestions

- Different output:
install/activate/update DCAT extension
- The widely used CKAN extension lacks mappings for popular extra keys (e.g. *spatial*)
- DCAT recommendation could provide properties for frequent extra keys, e.g., harvested datasets (`owl:sameAs`?)
- Distribution level should recommend(/allow) other dimension descriptions (e.g., `dct:temporal`, `dct:spatial`)

